Energy & Buildings 373 (2026) 118388

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild

Short-term cooling-load forecasting and chiller capacity recommendation with physics-enhanced residual-change LSTM

Zhenghao Gao [a], Nan Wang [a], Tong Wu [b,*], Shang Yang [a], Jian Xiao [a], Lunshu Chen [c], Junyao Feng [c]

[a] Guizhou Power Grid Co., Ltd., Guiyang, Guizhou, China
[b] University of Macau Zhuhai UM Science and Technology Research Institute, Zhuhai, Guangdong, China
[c] University of Macau, Macao, China

ARTICLE INFO

Keywords:
Cooling load forecasting
Chiller capacity matching
Equivalent thermal parameter
Residual-change LSTM
Building management system
Short-term operational decision support

ABSTRACT

Buildings contribute substantially to global carbon emissions, and chiller plants are major electricity consumers in large public buildings. Cooling demand varies continuously, whereas chillers operate in discrete combinations of rated capacities. Accurate forecasting can improve capacity matching by integrating operating information with building thermal response. This study develops a decision-support framework combining an equivalent thermal parameter (ETP) model, a Physics-Enhanced residual-change long short-term memory (LSTM) network, and a capacity recommendation method. The ETP model constructs a cooling-load baseline from weather, simplified envelope dynamics, and a training-selected chiller schedule, with thermal-response coefficients calibrated using measured loads. The LSTM combines historical building management system (BMS) records with ETP-related information to predict changes relative to the latest observed residual and reconstruct the next-step load. Under the fixed within-month partition and historical-weather reference, five independent training runs achieved a mean 10-min-ahead RMSE of 117.3 kW on 6699 common test windows, improving on a conventional LSTM and a benchmark that carries the latest measured baseline error forward. The forecast was adjusted by a validation-selected margin and mapped to the minimum feasible chiller-capacity combination. Under the same evaluation setting, mean recommended-capacity coverage across the five runs was 97.8% during public opening hours over the 56-day test set.

1. Introduction

Buildings account for a substantial share of global energy use and carbon emissions [1], with heating, ventilation, and air-conditioning (HVAC) systems contributing substantially in large public facilities [2]. Cooling loads vary with weather, occupancy, operating schedules, and envelope thermal response [3]. Weather conditions can also increase cooling demand and reduce air-conditioning efficiency, amplifying peak electricity consumption [4]. Central chiller plants meet this continuous demand through discrete on/off combinations, so a small load increase may require a larger increase in active capacity. Conventional threshold-based controls mainly respond to current conditions [5], making capacity matching sensitive to load changes. Research on optimized HVAC control [6] and robust chiller sequencing under measurement uncertainty [7] motivates the use of short-term forecasts to support capacity decisions.

Physics-based models provide a structured basis for cooling-load forecasting. Zhang et al. developed a climate-aware building energy model incorporating ambient-temperature and solar-radiation effects on cooling demand, air-conditioner COP, and electricity consumption [8]. Song et al. used a simplified equivalent thermal parameter (ETP) model to connect building thermal dynamics with indoor temperature, outdoor conditions, and air-conditioning power [9]. Qi et al. similarly employed a first-order ETP model to represent cooling demand and HVAC power consumption [10]. These studies demonstrate the usefulness of ETP models for representing thermal response with simplified building descriptions. However, calibrated baselines can retain time-varying errors arising from approximate building parameters and changing operating conditions.

Long short-term memory (LSTM) networks capture nonlinear temporal dependencies and are widely used for building cooling-load and energy forecasting [11,12]. Jiang et al. combined LSTM with

* Corresponding author.
E-mail addresses: 30065607@qq.com (Z. Gao), 764657828@qq.com (N. Wang), mc14944@um.edu.mo (T. Wu), yangshang2025@163.com (S. Yang), xiaojian_hrbeu@163.com (J. Xiao), mc25168@um.edu.mo (L. Chen), mc35156@um.edu.mo (J. Feng).

https://doi.org/10.1016/j.enbuild.2026.118388
Received 11 August 2026; Received in revised form 18 September 2026; Accepted 11 October 2026
Available online 14 October 2026
0378-7788/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

conformalized quantile regression and ensemble learning for HVAC energy prediction in public buildings [13]. Wang et al. applied a quantile-regression-based LSTM to probabilistic individual energy-consumption forecasting [14]. These approaches demonstrate the value of learning temporal patterns and prediction uncertainty. Their forecasting outputs nevertheless require an additional decision method to identify suitable chiller-capacity combinations.

Hybrid models combine physical structure with data-driven correction [15]. Interpretable neural architectures have also been developed for cooling-load forecasting [16]. This integration offers a way to retain a thermal reference while learning deviations associated with actual operation. Detailed chiller optimization can translate forecasts into operating schedules, but commonly requires equipment performance curves, auxiliary-power models, and operating constraints that may be incompletely documented in existing facilities. This motivates a forecasting-to-recommendation method that uses available operating records and simplified thermal information to connect predicted demand with discrete chiller capacities.

ARIMA models describe autoregressive and moving-average structure in load histories, whereas conventional LSTMs learn a direct mapping from historical inputs to load. Absolute-residual learning predicts the discrepancy from a physical baseline. The present formulation instead predicts changes from the latest observed residual and combines them with the target-time ETP baseline.

Two linked challenges are therefore addressed in this study: correcting errors in a calibrated ETP baseline as operating conditions change, and translating continuous load forecasts into discrete capacity recommendations. A schedule-modulated ETP model constructs a served-load baseline from weather, simplified envelope thermal response, and a training-selected chiller schedule [17]. The Physics-Enhanced residual-change LSTM combines full BMS histories with ETP-related information to predict changes relative to the latest observed baseline residual. The target-time ETP baseline, current measured residual, and learned residual change jointly reconstruct the next-step cooling-load forecast.

The resulting 10-min-ahead forecast is adjusted using a validation-calibrated margin and mapped to the minimum feasible chiller combination. Schedule-aware low-load rules account for inactive periods, while capacity coverage, excess capacity, shortfall, and recommendation changes characterize the resulting decisions.

The main contributions of this study are as follows:

1) A schedule-modulated ETP baseline combines available building information with approximate thermal parameters. Its response coefficients and operating schedule are determined using training data, providing a thermal reference without requiring detailed construction information.
2) A Physics-Enhanced residual-change LSTM integrates full BMS histories with ETP-related information. The ETP baseline participates in both residual definition and forecast reconstruction, combining physical response with the latest measured operating state to improve next-step cooling-load prediction.
3) A forecasting-to-recommendation method maps the adjusted 10-min-ahead forecast to the minimum feasible combination of unequal-sized chillers. Validation-based margin selection and schedule-aware low-load rules balance capacity coverage and unnecessary active capacity.

The framework is evaluated through forecasting accuracy, capacity adequacy, recommendation changes, and daily rated-capacity allocation. Section 2 presents the forecasting method, and Section 3 describes capacity recommendation, followed by the experimental evaluation.

2. Physics-enhanced short-term cooling-load forecasting

Fig. 1 outlines the forecasting-to-capacity framework. The LSTM uses historical operating and ETP-related variables to predict residual changes. These predictions are combined with the latest observed residual and the target-time ETP baseline to reconstruct cooling load. The 10-min-ahead load forecast is then used to recommend a chiller combination.

Table 1
Equivalent building and envelope assumptions used in the ETP model.

Parameter	Value
Total floor area, A	32 245 m^2
Above-ground floors, N_f	5
Equivalent floor height, h_f	4.0 m
Window-to-wall ratio, WWR	0.35
Wall U-value, U_{wall}	1.20 W/(m^2 K)
Roof U-value, U_{roof}	0.80 W/(m^2 K)
Window U-value, U_{win}	3.30 W/(m^2 K)
Solar heat-gain coefficient, SHGC	0.45
Shading factor, f_{shade}	0.80
Solar absorptance, α_{sol}	0.60
External convection coefficient, h_{ext}	15 W/(m^2 K)
Effective areal heat capacity, c_{eff}	165 kJ/(m^2 K)

2.1. Modelling of cooling chiller operation

The cooling rate was calculated from the chilled-water mass flow rate and the return-to-supply temperature difference using the liquid-side capacity relation [18]:

$$Q_t^{\mathrm{water}} = \dot{m}_{w,t} c_p \left(T_{w,t}^{\mathrm{ret}} - T_{w,t}^{\mathrm{sup}} \right), \tag{1}$$

where Q_t^{water} is the water-side cooling rate at time t, $\dot{m}_{w,t}$ is the chilled-water mass flow rate, $c_p = 4.203$ kJ/(kg · °C) is the specific heat capacity of water, and $T_{w,t}^{\mathrm{ret}}$ and $T_{w,t}^{\mathrm{sup}}$ are the return and supply-water temperatures. The resulting cooling energy for each 10-min interval was stored in the BMS dataset. In the present preprocessing pipeline, the recorded interval energy $E_{\mathrm{cool},t}$ was converted to the corresponding average served cooling load as:

$$Q_t^{\mathrm{meas}} = \frac{E_{\mathrm{cool},t}}{\Delta t} = 6 E_{\mathrm{cool},t}, \qquad \Delta t = \frac{10}{60}\ \mathrm{h}, \tag{2}$$

where Q_t^{meas} is the average served cooling load in kW, $E_{\mathrm{cool},t}$ is the interval cooling energy, and Δt is the 10-min interval duration in hours. This target represents the cooling load delivered under historical BMS operation.

Cooling-load values from 0% to 110% of the installed capacity were used for model development. The range removes negative and unusually high readings while allowing a 10% tolerance for short-term fluctuations and measurement errors. Records outside this range remained in the chronological data splits but were excluded from ETP calibration and forecasting-window generation.

2.2. Forecasting task and overall framework

The model uses six consecutive 10-min observations to predict 18 future interval-average water-side loads. Fig. 2 shows the ETP-assisted residual-change architecture; its first output supplies the 10-min-ahead forecast evaluated and used for capacity recommendation.

2.2.1. Weather and continuous thermal state

Weather data were mapped to the 10-min BMS timestamps. Outdoor temperature, humidity, radiation, wind, cloud cover, and precipitation provide the time-varying forcing. Moist-air enthalpy was calculated from temperature and humidity; the psychrometric expressions and constants are given in Supplementary Section S1.

The indoor reference enthalpy h_{in} was calculated using assumed indoor conditions of 24 °C and 55% relative humidity. The building is represented as a single equivalent thermal zone using a first-order ETP model [19]. Table 1 summarizes the engineering assumptions used to

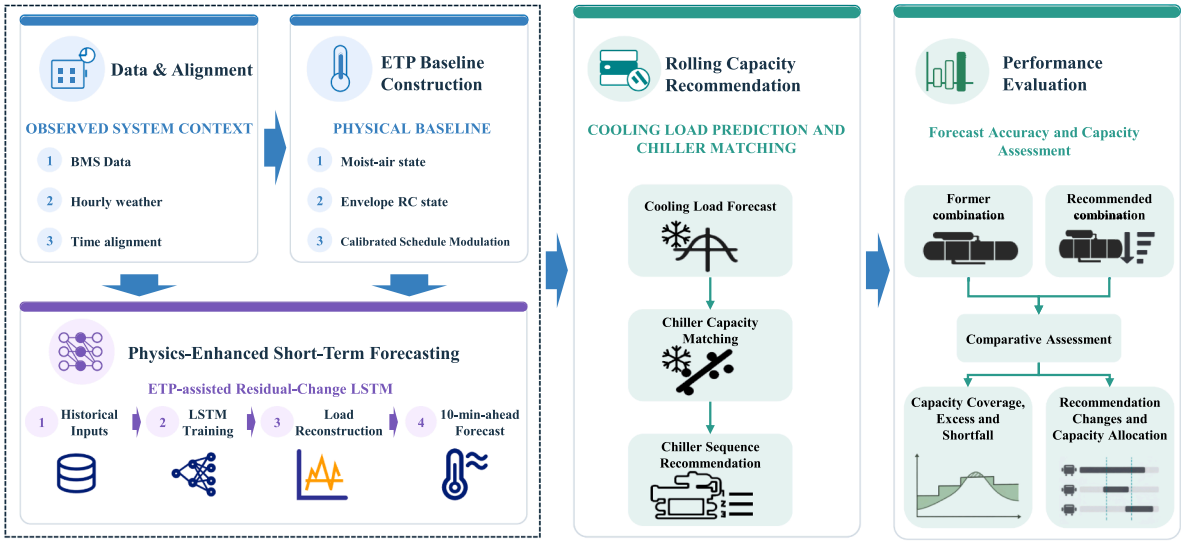


Fig. 1. Overall framework for physics-enhanced short-term cooling-load forecasting and chiller-capacity recommendation.

estimate the opaque-wall, window, and roof areas and thermal conductances. Because complete construction drawings and material records were unavailable, typical values were assigned to the unavailable thermal parameters. The optimized thermal-response coefficients were then calibrated by measured cooling-load data.

Envelope areas were estimated assuming a square building footprint. The geometric expressions and parameter definitions are given in Supplementary Section S1. Static envelope properties determine heat-transfer conductances and effective thermal capacitance; weather and the preceding thermal state produce their dynamic response.

A solar-air temperature represents shortwave heating of the opaque exterior surface:

$$T_{\mathrm{sa},t} = T_{\mathrm{out},t} + \frac{\alpha_{\mathrm{sol}} I_t^+}{h_{\mathrm{ext}}}, \qquad I_t^+ = \max(I_t, 0). \tag{3}$$

where $T_{\mathrm{sa},t}$ is the solar-air temperature, $T_{\mathrm{out},t}$ is the outdoor dry-bulb temperature, and I_t is the interpolated shortwave irradiance in $\mathrm{W/m^2}$. The opaque-envelope thermal state is propagated as:

$$\begin{aligned} C_{\mathrm{eff}} &= c_{\mathrm{eff}} A, \\ \tau &= \frac{1000 C_{\mathrm{eff}}}{3600 U A_{\mathrm{opaque}}}, \\ \phi_t &= \exp\left(-\frac{\Delta t_t}{\tau}\right), \\ T_{\mathrm{node},t} &= \phi_t T_{\mathrm{node},t-1} + (1-\phi_t) T_{\mathrm{sa},t} \end{aligned} \tag{4}$$

where C_{eff} is the effective thermal capacitance in kJ/K, UA_{opaque} is the combined wall-and-roof conductance in W/K, τ is the equivalent time constant in hours, Δt_t is the elapsed time in hours, ϕ_t is the state-retention factor, and $T_{\mathrm{node},t}$ is the opaque-envelope node temperature. The factors 1000 and 3600 convert the resulting time constant to hours. Using the equivalent areas in Supplementary Eq. (S2), the opaque-envelope conductance is $UA_{\mathrm{opaque}} = U_{\mathrm{wall}} A_{\mathrm{wall}} + U_{\mathrm{roof}} A_{\mathrm{roof}}$.

The resulting non-negative envelope contribution is:

$$Q_{\mathrm{env},t} = \max\left\{ \frac{UA_{\mathrm{opaque}}(T_{\mathrm{node},t} - T_{\mathrm{in}})}{1000} + \frac{UA_{\mathrm{win}}(T_{\mathrm{out},t} - T_{\mathrm{in}})}{1000} + \frac{A_{\mathrm{win}} \mathrm{SHGC} f_{\mathrm{shade}} I_t^+}{1000}, 0 \right\} \tag{5}$$

where $Q_{\mathrm{env},t}$ is the envelope cooling-load contribution in kW, $T_{\mathrm{in}} = 24\,^{\circ}\mathrm{C}$ is the indoor reference temperature, A_{win} is the equivalent window area, and $UA_{\mathrm{win}} = U_{\mathrm{win}} A_{\mathrm{win}}$ is the window conductance. The three terms represent opaque conduction, window conduction, and window solar gain, respectively.

2.2.2. Training-selected chiller schedule

The public-opening indicator o_t equals 1 between 10:00 and 18:00 on operating days and 0 otherwise. It represents public access, whereas the chiller schedule may include pre-cooling before opening and gradual-shutdown after closing. The reference step schedule therefore assigns a weight of 1 during public-opening hours and 0.05 at other times.

To represent the daily chiller schedule more flexibly, a set of trapezoidal profiles was evaluated. Let m_t denote the minute of day at time t. The schedule weight is defined as:

$$\begin{aligned} u_t &= \mathrm{clip}\left(\frac{m_t - m_{\mathrm{up,start}}}{m_{\mathrm{up,end}} - m_{\mathrm{up,start}}}, 0, 1\right), \\ v_t &= \mathrm{clip}\left(\frac{m_{\mathrm{down,end}} - m_t}{m_{\mathrm{down,end}} - m_{\mathrm{down,start}}}, 0, 1\right), \\ w_t &= c + (1-c)\min(u_t, v_t) \end{aligned} \tag{6}$$

where $\mathrm{clip}(x, a, b)$ limits x to $[a, b]$. The start-up factor u_t rises from 0 to 1 between $m_{\mathrm{up,start}}$ and $m_{\mathrm{up,end}}$; the shutdown factor v_t falls from 1 to 0 between $m_{\mathrm{down,start}}$ and $m_{\mathrm{down,end}}$. The minimum close-hour weight is c, giving $c \le w_t \le 1$.

The plant schedule was selected by four-fold cross-validation using only the training days. The training days from each month were distributed among four folds, and each day was assigned to one fold. Candidate trapezoidal profiles used $c \in \{0, 0.05, 0.10, 0.15\}$. Ramp-up started at 07:30, 08:00, or 08:30 and lasted 60, 90, or 120 min, with completion by 10:00. Ramp-down started at 18:00, 18:30, or 19:00 and lasted 90, 120, or 150 min. These settings produced 288 valid profiles.

For each candidate, ETP coefficients were fitted on three folds and evaluated on the fourth. The lowest mean held-out RMSE selected $c = 0.05$, an 08:00–09:00 start-up ramp, and an 18:00–20:30 shutdown ramp; this schedule was fixed for validation and testing.

2.2.3. Model calibration

Positive dry-bulb and enthalpy deviations from the indoor reference are defined as:

$\Delta T_t^+ = \max(T_{\mathrm{out},t} - T_{\mathrm{in}}, 0)$ and $\Delta h_t^+ = \max(h_{\mathrm{out},t} - h_{\mathrm{in}}, 0)$.

The unmodulated specific-load model is:

$$q_{\mathrm{phys},t} = k_{\mathrm{env}} \frac{Q_{\mathrm{env},t}}{A} + k_T \Delta T_t^+ + k_h \Delta h_t^+ + k_s I_t^+ + q_{\mathrm{base}} + q_{\mathrm{open}} o_t \tag{7}$$

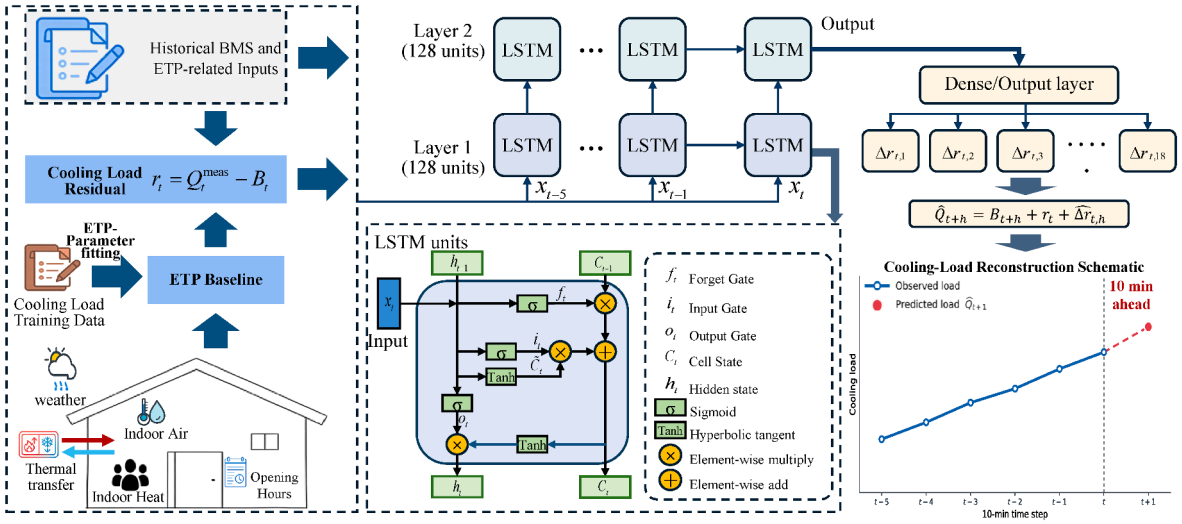


Fig. 2. LSTM and vector-decoder layout for multi-step served cooling-load forecasting.

where $q_{\mathrm{phys},t}$ is the estimated cooling load per unit floor area in kW/m^2; k_{env} scales the envelope term, while k_T, k_h, and k_s provide empirical corrections associated with dry-bulb deviation, enthalpy deviation, and solar irradiance to improve agreement with the measured served cooling load; and q_{base} and q_{open} are the background and public-opening terms.

The schedule-modulated ETP baseline is:

$$B_t = Q_{\mathrm{ETP},t} = A \max\left(w_t q_{\mathrm{phys},t}, 0\right), \tag{8}$$

where B_t and $Q_{\mathrm{ETP},t}$ denote the same ETP served-load baseline in kW.

The calibration set includes valid training samples from public-opening hours:

$$\begin{aligned} &\mathcal{T}_{\mathrm{cal}} = \{t \in \mathcal{T}_{\mathrm{train}} : o_t = 1, \quad 0 < Q_t^{\mathrm{meas}} \le Q_{\max}\}, \\ &Q_{\max} = 1.1 C_{\mathrm{total}} = 3636.6 \text{ kW}. \end{aligned} \tag{9}$$

where $\mathcal{T}_{\mathrm{train}}$ is the set of training timestamps, $\mathcal{T}_{\mathrm{cal}}$ is the calibration subset, $C_{\mathrm{total}} = 3306$ kW is the total chiller capacity, and $Q_{\max}$ is the upper cooling load threshold. Because $o_t = 1$ for all calibration samples, q_{base} and q_{open} could not be estimated separately. Therefore, q_{open} was set to zero, and its contribution was included in q_{base}.

The parameter vector is estimated by:

$$\theta^* = \arg\min_{\theta \in \Theta} \left[\frac{1}{|\mathcal{T}_{\mathrm{cal}}|} \sum_{t \in \mathcal{T}_{\mathrm{cal}}} \left(B_t - Q_t^{\mathrm{meas}}\right)^2 + \lambda_\theta \|\theta\|_2^2 \right]. \tag{10}$$

$$\Theta = \left\{ \theta : \begin{array}{l} k_{\mathrm{env}} \ge 0.1, \quad q_{\mathrm{open}} = 0, \\ k_T, k_h, k_s, q_{\mathrm{base}} \ge 0 \end{array} \right\} \tag{11}$$

where $\theta = [k_{\mathrm{env}}, k_T, k_h, k_s, q_{\mathrm{base}}, q_{\mathrm{open}}]^{\mathrm{T}}$ is the coefficient vector, and θ^* is its fitted value. The terms $|\mathcal{T}_{\mathrm{cal}}|$ and $\lambda_\theta = 10^{-4}$ denote the number of calibration observations and the regularization coefficient, respectively.

The non-negative basis terms, schedule weights, and feasible coefficients make B_t linear in θ, allowing bounded ridge least squares. For numerical stability, the ridge-augmented design matrix was scaled by column Euclidean norms, with bounds transformed and fitted coefficients restored to their original units, preserving the ridge penalty.

2.3. Physics-enhanced LSTM forecasting

2.3.1. Input and forecast windows

For each forecast origin t, the historical input and multi-horizon target are defined as:

$$\begin{aligned} &\mathbf{X}_t = [\mathbf{x}_{t-5}, \ldots, \mathbf{x}_t] \in \mathbb{R}^{6 \times p}, \\ &\mathbf{y}_t = [Q_{t+1}^{\mathrm{meas}}, \ldots, Q_{t+18}^{\mathrm{meas}}] \in \mathbb{R}^{18} \end{aligned} \tag{12}$$

where $\mathbf{x}_t$ is the p-dimensional feature vector, $\mathbf{X}_t$ contains six consecutive 10-min samples, and $\mathbf{y}_t$ contains the served cooling loads from 10 to 180 min ahead. The input therefore corresponds to a nominal 60-min historical window.

Forecast windows were generated within each complete day. Each window required continuous 10-min timestamps and valid input and target values. Windows crossing midnight or a data-split boundary were excluded.

The historical input is the ordered union of the original residual-model variables and the BMS features used by the conventional LSTM, giving $p = 145$ and $\mathbf{X}_t \in \mathbb{R}^{6 \times 145}$. These inputs include operating measurements, weather, enthalpy, calendar and schedule descriptors, and the historical ETP baseline. Historical cooling-energy measurements are retained as inputs. Future load labels and recommendation-derived variables are excluded. The target-time ETP baseline enters load reconstruction, as shown in Fig. 1. The complete historical input inventory is provided in Supplementary Section S9 and Table S10.

2.3.2. Residual-change forecasting

Residual learning has been applied to building-energy prediction [20]. In this study, the residual between the measured served load and the ETP baseline is defined as:

$$\begin{aligned} &r_\tau = Q_\tau^{\mathrm{meas}} - B_\tau, \\ &\Delta r_{t,h} = r_{t+h} - r_t, \qquad h = 1, \ldots, 18, \\ &\hat{Q}_{t+h}^{\mathrm{hyb}} = B_{t+h} + r_t + \widehat{\Delta r}_{t,h} \end{aligned} \tag{13}$$

Here t and h denote the forecast origin and step, r_τ is the ETP residual at timestamp τ, and $\widehat{\Delta r}_{t,h}$ predicts its change from the current residual r_t.

During training, $\Delta r_{t,h}$ is calculated from the measured residuals at the forecast origin and target timestamp. During inference, the LSTM predicts residual changes from the historical input sequence. Load reconstruction uses the current observed residual r_t , calculated from the latest measured load and the corresponding ETP baseline; future measured residuals are not required. Windows are retained when r_t is finite and the latest measured load is within $[0, Q_{\max}]$.

The ETP baseline supplies the target-time thermal reference, the current residual aligns it with the observed operating state, and the learned change updates the forecast.

For comparison, a latest-residual persistence baseline assumes no change in the residual:

$$\hat{Q}_{t+h}^{\mathrm{pers}} = B_{t+h} + r_t, \tag{14}$$

This benchmark holds the current residual constant while sharing the target-time ETP baseline with the Physics-Enhanced model.

2.3.3. Model training and forecast reconstruction

Table 2 gives the neural architectures and training settings. The residual model's prediction head applies LayerNorm, a linear layer, ReLU, dropout, and a final linear layer to generate the 18 residual-change outputs.

Input variables and residual-change targets were standardized using training statistics only (Supplementary Section S1). Training minimizes the mean squared error over the 18 jointly predicted residual changes:

$$\mathcal{L} = \frac{1}{N_{\mathrm{b}} N_{\mathrm{out}}} \sum_{n=1}^{N_{\mathrm{b}}} \sum_{h=1}^{N_{\mathrm{out}}} \left(\tilde{z}_{n,h} - \hat{\tilde{z}}_{n,h} \right)^2, \tag{15}$$

Here $\mathcal{L}$ is the minibatch loss, N_{b} is batch size, $N_{\mathrm{out}} = 18$, and $\tilde{z}_{n,h}$ and $\hat{\tilde{z}}_{n,h}$ are the standardized target and prediction. The checkpoint with minimum validation loss is restored; training stops after 20 epochs without improvement.

For the reference evaluation, B_{t+h} was generated using historical Open-Meteo weather at the target timestamps. Weather and BMS records were aligned to the same local time axis. The primary forecasting comparison uses this historical-weather reference; a separate archived-forecast evaluation reports the effect of replacing target-time weather. In operational use, the target-time weather input is supplied by forecasts issued at or before the decision origin [21].

All methods are compared on the same 6699 windows using first-output RMSE, MAE, and Bias (Supplementary Section S1). Training retains the joint 18-output loss; evaluation and current capacity recommendation use the 10-min-ahead output.

3. Chiller capacity recommendation

Short-term load forecasts can support chiller sequencing decisions [22]. The proposed method updates the cooling-load forecast and capacity recommendation every 10 min. The next-step forecast is adjusted using validation underprediction errors and processed by a schedule-aware rule. The resulting requirement is mapped to the minimum feasible chiller combination.

3.1. Observed state and candidate capacities

Historical BMS states define the active chiller combination and rated capacity:

$$\begin{aligned} & z_{i,t}^{\mathrm{BMS}} \in \{0,1\}, \\ & S_t^{\mathrm{BMS}} = \left\{ i \mid z_{i,t}^{\mathrm{BMS}} = 1 \right\}, \\ & C_t^{\mathrm{BMS}} = \sum_{i=1}^{3} z_{i,t}^{\mathrm{BMS}} C_i \end{aligned} \tag{16}$$

Here $z_{i,t}^{\mathrm{BMS}}$ is the recorded on/off state and C_i is chiller i's rated capacity. C_t^{BMS} is historical active capacity, whereas Q_t^{meas} is delivered cooling load.

All subsets of the three chillers, including the empty set, form the candidate combinations:

$$\mathcal{S} = 2^{\{1,2,3\}}, \qquad C(S) = \sum_{i \in S} C_i, \quad S \in \mathcal{S}. \tag{17}$$

where $\mathcal{S}$ is the candidate set, S is one chiller combination, and $C(S)$ is its total rated capacity. The three chillers provide six distinct capacity levels: 0, 842, 1,232, 2,074, 2,464, and 3,306 kW. Combinations with the same capacity are retained separately so that the previous combination can be maintained when possible.

3.2. Margin-adjusted 10-min-ahead load requirement

For capacity recommendation, $\hat{Q}$ denotes the reconstructed forecast clipped at zero; forecast-error metrics use the reconstruction before clipping. For validation sample j with origin t_j, define underprediction as $s_j = Q_{t_j+1}^{\mathrm{meas}} - \hat{Q}_{t_j+1}$. The one-sided margin and adjusted forecast are:

$$\begin{aligned} & m = \max\left(\mathrm{Quantile}_{0.93}\{s_j\}_{j=1}^{n_v},\ 0\right), \\ & \hat{Q}_{t+1}^{\mathrm{adj}} = \max\left(\hat{Q}_{t+1} + m,\ 0\right) \end{aligned} \tag{18}$$

where positive values of s_j represent underprediction and n_v is the number of valid validation samples. The empirical quantile uses the sorted error at rank $\min\{\lceil (n_v+1)\alpha \rceil, n_v\}$, with $\alpha = 0.93$ selected on the validation set. Candidate quantiles were 0.95, 0.94, 0.93, 0.925, 0.90, 0.875, and 0.85. Relative to the 0.95 reference, admissible candidates retained public-opening capacity coverage within 0.5 percentage points and limited the mean positive load shortfall to 1.2 times the reference. The candidate with the lowest mean positive excess capacity was selected, giving $m = 130.972$ kW from 6031 validation samples.

For the OFF rule, the normalized schedule is $f_{\mathrm{sch},t} = (w_t - c)/(1-c)$. Values 0 and 1 represent inactive and full scheduled plant operation, respectively, distinct from the public-opening indicator o_t.

A low-load threshold was used to determine whether all chillers could remain OFF during inactive periods. The threshold was calculated from training samples in which $f_{\mathrm{sch},t} = 0$ and all chillers were recorded as OFF:

$$\begin{aligned} & Q_{\mathrm{off}} = \max\left[Q_{\min},\ \mathrm{Quantile}_{0.99}\left\{ Q_t^{\mathrm{meas}} : t \in \mathcal{T}_{\mathrm{off}} \right\} \right], \\ & Q_{\min} = 50\ \mathrm{kW}. \end{aligned} \tag{19}$$

where $\mathcal{T}_{\mathrm{off}}$ contains the training timestamps at which $f_{\mathrm{sch},t} = 0$ and all chillers were recorded as OFF.

Short-term cooling-load forecasts support HVAC control, including chiller sequencing [23]. The schedule state and the raw and adjusted forecasts define the basic capacity requirement:

$$R_{t+1} = \begin{cases} 0, & f_{\mathrm{sch},t+1} = 0 \text{ and } \hat{Q}_{t+1} \le Q_{\mathrm{off}}, \\ & \text{or } \hat{Q}_{t+1}^{\mathrm{adj}} \le Q_{\min}, \\ \hat{Q}_{t+1}^{\mathrm{adj}}, & \text{otherwise.} \end{cases} \tag{20}$$

A post-closing rule takes precedence over Eq. (20). When the forecast origin is at or after 18:00, all chillers are observed OFF, and both the observed load and next-step raw forecast lie between zero and Q_{off}, the final requirement R_{t+1} is set to zero. This rule requires consecutive 10-min observations within the same day and uses the state and load available at the forecast origin. It allows an already stopped plant to remain OFF during low-load evening periods. Otherwise, the requirement from Eq. (20) is retained.

3.3. Minimum feasible capacity recommendation

The requirement R_{t+1} represents the cooling capacity needed at time $t+1$. If $R_{t+1} = 0$, the empty combination is recommended. For a positive requirement within the installed capacity, the method selects the smallest available capacity that can meet the requirement:

$$S_{t+1}^{\mathrm{rec}} \in \underset{S \in \mathcal{S}:\, C(S) \ge R_{t+1}}{\arg\min} C(S), \qquad 0 < R_{t+1} \le C_{\mathrm{total}}. \tag{21}$$

If multiple combinations provide the same minimum feasible capacity, the previous recommendation is retained when possible. Otherwise, the combination with the largest overlap with the previous recommendation is selected, and a fixed order resolves any remaining tie. This rule reduces unnecessary chiller switching. For an illustrative forecast of 1,100 kW, the selected margin gives a requirement of 1,230.972 kW, which can be met by one 1,232 kW chiller. A forecast of 1,150 kW gives 1,280.972 kW and selects the 2,074 kW level. Thus the margin affects recommendations chiefly near the discrete capacity boundaries.

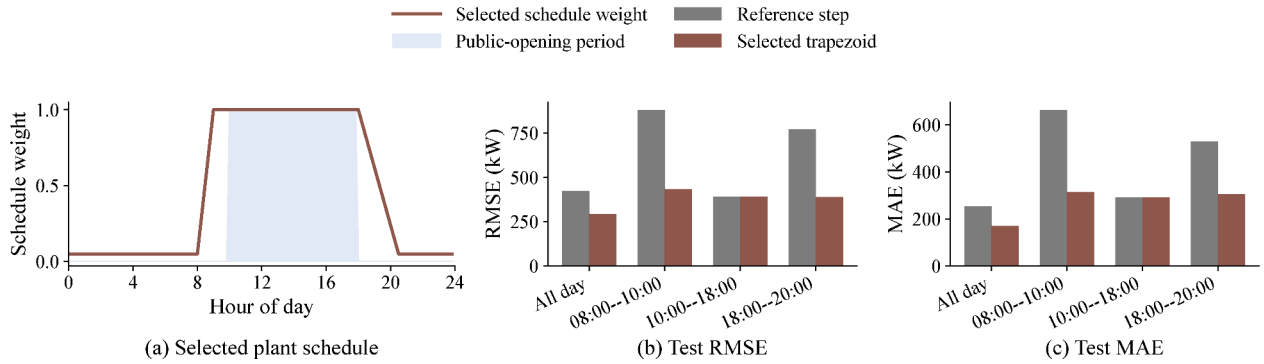


Fig. 3. Training-selected chiller schedule and test-set ETP schedule ablation. Panel (a) shows the selected daily modulation profile, while panels (b) and (c) compare the reference public-opening step and selected trapezoidal schedule using RMSE and MAE. The all-day ablation uses 8019 valid test records, whereas Fig. 5 and Supplementary Table S1 use 6699 common forecast windows.

Each test day is an independent episode initialized from the latest BMS combination observed at or before its first decision. Recommendations begin with the first valid forecast. The previous recommended chiller combination is not carried over between test dates.

If $R_{t+1} > C_{\text{total}}$, all chillers are selected and the infeasible requirement is recorded as $d_{t+1}^{\text{req}} = \max(R_{t+1} - C_{\text{total}}, 0)$.

The recommendation identifies the minimum feasible rated capacity at each update; equipment-specific start-up and minimum-runtime constraints belong to the subsequent plant-control implementation.

3.4. Capacity-adequacy and switching diagnostics

The evaluation set $\mathcal{T}_{\text{eval}}$ contains timestamps with a valid recommendation and a valid measured load. For policy-adjusted evaluation, the comparison load Q_{t+1}^{cmp} is set to zero when a policy-defined OFF rule is triggered; otherwise, it equals Q_{t+1}^{meas}. Coverage against the unadjusted measured load is reported separately.

For an evaluation subset $\mathcal{T}$ and a stated comparison requirement q_{t+1}, capacity coverage and shortfall are defined as:

$$\begin{aligned} \text{Cov}(\mathcal{T}, q) &= \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbb{1}\left[C(S_{t+1}^{\text{rec}}) \geq q_{t+1}\right], \\ \ell_t(q) &= \max\left[q_{t+1} - C(S_{t+1}^{\text{rec}}), 0\right]. \end{aligned} \tag{22}$$

Capacity coverage is reported separately for all valid measured targets, public opening hours ($o_{t+1} = 1$), and scheduled plant operation ($f_{\text{sch},t+1} > 0$). Scheduled operation includes morning preparation and part of the evening ramp-down. Policy-adjusted coverage additionally sets the comparison load to zero for explicit OFF decisions. Mean excess capacity and mean shortfall are the positive capacity-minus-load and load-minus-capacity differences, respectively, averaged over every valid record in the stated subset. The excess-tier rate is the fraction of recommendations above the minimum discrete capacity required by the measured load.

A recommendation change is counted when $S_{t+1}^{\text{rec}} \neq S_t^{\text{rec}}$. One change may involve multiple chillers.

3.5. Capacity-configuration assessment

Capacity allocation is evaluated through the magnitude and duration of changes in recommended rated capacity. The comparison uses the policy-adjusted load Q_{t+1}^{cmp} defined in Section 3.4. Comparable intervals are valid records at which both historical and recommended capacities meet this requirement. For these intervals, define $\Delta C_t = C_{t+1}^{\text{BMS}} - C(S_{t+1}^{\text{rec}})$. Positive values represent downshifts and negative values represent upshifts. The daily integrals are:

$$\begin{aligned} I_d^{\downarrow} &= \sum_{t \in \mathcal{T}_{\text{cmp},d}} [\Delta C_t]_+ \Delta t, \\ I_d^{\uparrow} &= \sum_{t \in \mathcal{T}_{\text{cmp},d}} [-\Delta C_t]_+ \Delta t, \\ I_d^{\text{net}} &= I_d^{\downarrow} - I_d^{\uparrow}, \quad \Delta t = 1/6 \text{ h}. \end{aligned} \tag{23}$$

where $[x]_+ = \max(x, 0)$, and $\mathcal{T}_{\text{cmp},d}$ contains the comparable intervals on day d. The integrals are calculated in kW·h of rated capacity and divided by 1000 for reporting in MW·h. They summarize the extent and duration of capacity reallocation. Raw-load coverage and shortfall accompany the schedule-adjusted comparison. Daily results include all 56 test days.

4. Experimental results and discussion

4.1. Case study and experimental setup

The case study covers a 32,245 m^2 public building complex in a coastal subtropical region, with exhibition, convention, and public-activity spaces. Cooling demand varies with weather, occupancy, event schedules, and thermal response, particularly during start-up and daytime operation.

The plant comprises three water-cooled chillers: CH-1 and CH-3 at 1,232 kW each, and CH-2 at 842 kW, totaling 3,306 kW. Their BMS on/off states and rated capacities define the historical capacity benchmark.

Chilled-water measurements provide the served cooling load, while the three chiller electricity meters provide the total electrical power. Fig. 4 shows a strong positive association between these quantities, with Pearson's $r = 0.920$ and $R^2 = 0.847$. This association supports the operational relevance of cooling-load forecasting.

Open-Meteo historical weather [24] supplied temperature, relative humidity, dew point, solar radiation, wind speed, cloud cover, and precipitation. Records were aligned with local BMS time and interpolated to 10-min intervals; temperature and humidity also supplied outdoor enthalpy.

The public-opening hours were 10:00–18:00, and Thursday was the regular closed day. The ETP envelope state was propagated through the complete chronological record to preserve thermal continuity. Closed days were then excluded, while both public-opening and non-public-opening periods on the retained dates were preserved.

The retained dates were partitioned within each calendar month using a fixed random seed of 42, with each complete date assigned to one subset and the resulting split held constant across all training runs. This produced 192 training days, 44 validation days, and 56 test days. Training data were used for ETP calibration, schedule selection, scaling parameters, and LSTM fitting. Validation data were used for checkpoint selection and margin calibration. All model configurations were evaluated on the same test dates. This design represents seasonal conditions

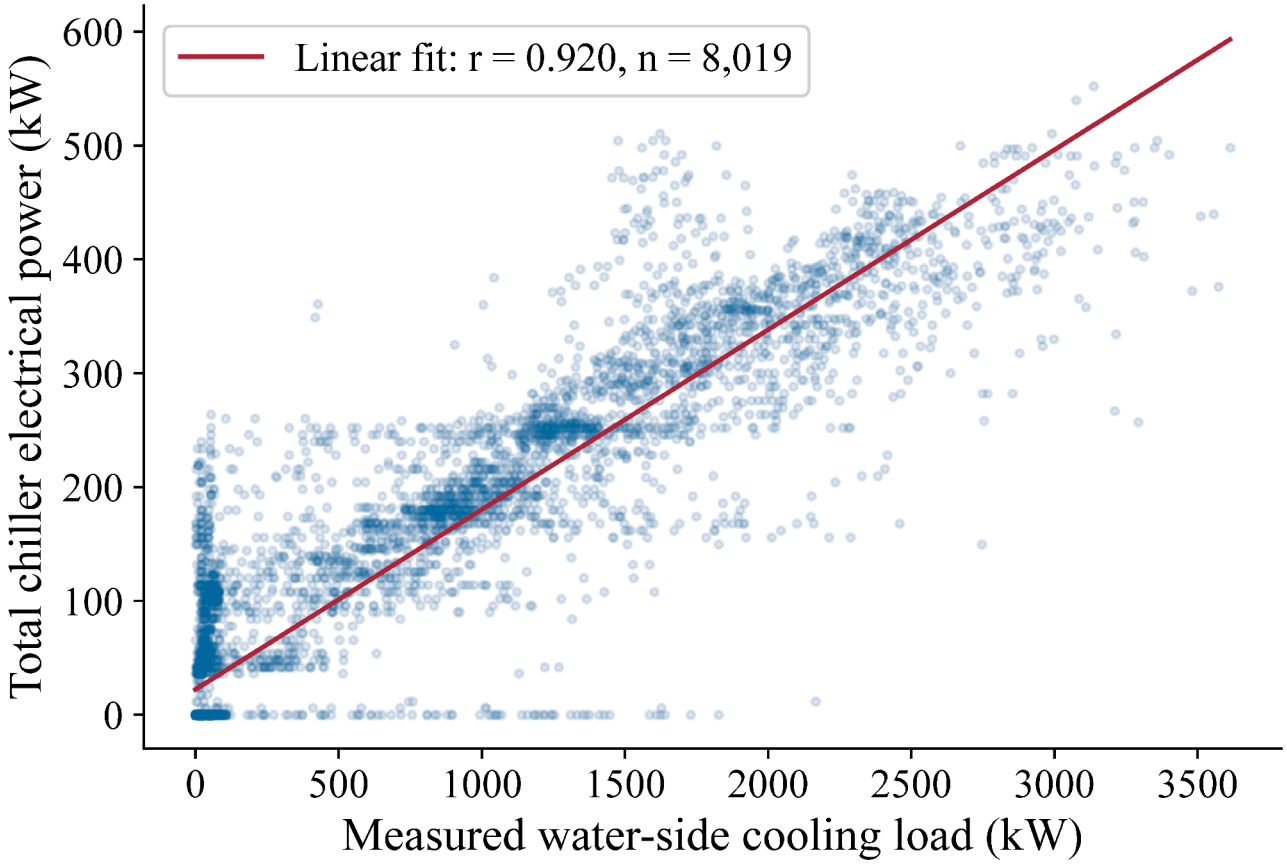


Fig. 4. Relationship between measured cooling load and chiller electrical power across 8019 test records, with a linear regression fit.

in each subset and evaluates interpolation among observed operating days. ETP state propagation across subset boundaries uses chronological weather forcing and the preceding thermal state; cooling-load measurements enter only calibration on training days and the observed histories available at each forecast origin.

Chronological expanding-window tests covered July, September, and November 2024, with the preceding month used for validation and earlier dates for training. ETP calibration, schedule selection, normalization, and network fitting were repeated within each training period, as detailed in Supplementary Section S7 and Tables S7– S8.

The forecasting comparison uses 6699 common windows with complete outputs across all models. The capacity analysis uses 7697 valid next-step recommendations, including late-day origins with a valid first output; 7691 of these have a valid measured next-step load. Forecasting windows remain within individual dates.

ETP calibration yielded $k_{\mathrm{env}} = 0.1$, $k_T = 0.003857139$, $k_h = 0.000881288$, $k_s = 0$, and $q_{\mathrm{base}} = 0.006622735$, with $q_{\mathrm{open}} = 0$ fixed for identifiability. The coefficients use the units implied by Eq. (7). The envelope coefficient reached its prescribed positive lower bound of 0.1, and the additional solar coefficient fitted to zero. These coefficients represent effective responses of the simplified baseline to the measured water-side load. The temperature and enthalpy terms capture weather-related variation, while schedule modulation represents the recurring plant operating pattern. Solar forcing also enters the envelope node temperature and window-gain term, so its role is retained through the physical baseline.

Table 2
Neural architectures and shared training settings.

Item	Physics-Enhanced	Conventional LSTM
Input features	145	131
Input shape	6×145	6×131
LSTM layers	2	3
Outputs	18	18
Shared training settings		
Hidden units	128	
Dropout	0.20	
Optimizer / LR	AdamW / 0.001	
Batch size	128	
Epochs / patience	100 / 20	

Fig. 3 shows the selected trapezoidal schedule and its ETP ablation against the public-opening step schedule. Lower RMSE and MAE demonstrate the benefit of representing start-up and shutdown transitions.

4.2. Forecasting and capacity recommendation performance

Each model was trained in five independent runs using random seeds 42–46, with matching seeds used for the two model families. All runs used the fixed configurations in Table 2, the same data split, training-only scaling, and validation-based checkpoint selection. The Physics-Enhanced run with random seed 42 serves as the reference run for detailed capacity, sensitivity, and weather analyses; it completed 72

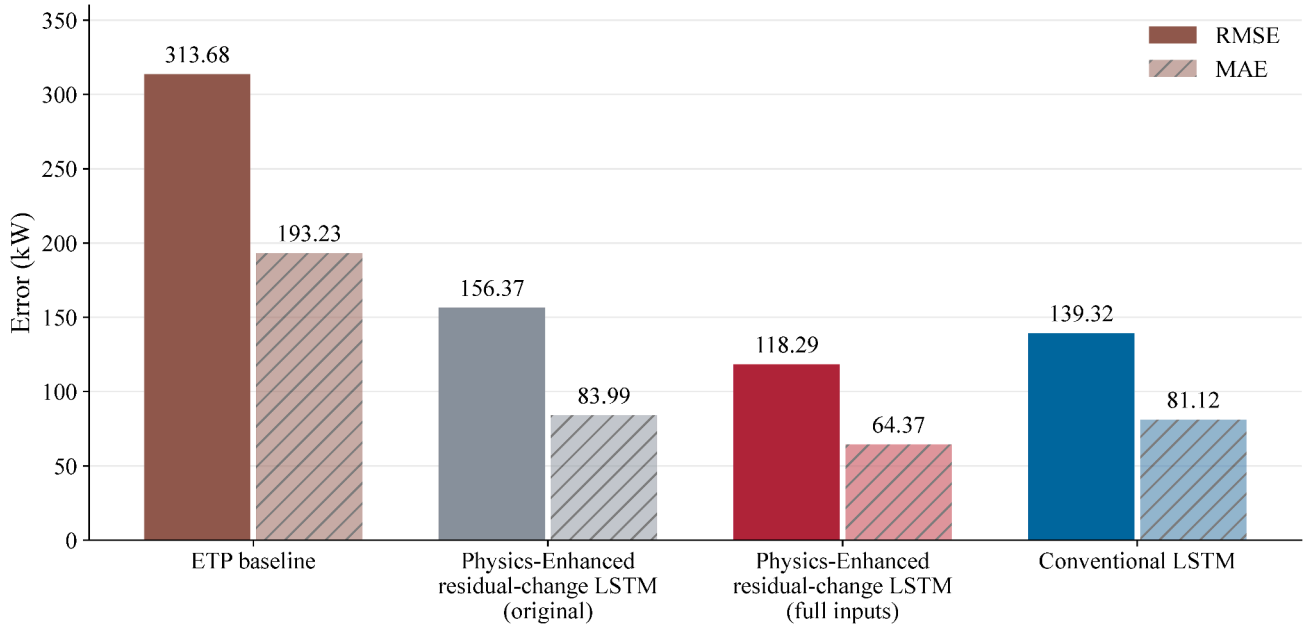


Fig. 5. Reference-run 10-min-ahead RMSE and MAE on 6699 common test windows. The original and full-input residual models use 27 and 145 historical variables, respectively.

epochs, with the checkpoint from epoch 52 selected. Supplementary Section S5 and Table S3 compare one to three recurrent layers across five runs per configuration. Table S4 further shows that the one-layer models selected by mean validation RMSE retained Physics-Enhanced's advantage, reducing mean RMSE and MAE by 12.58% and 18.13%, respectively.

Fig. 5 shows reference-run RMSE and MAE reductions of 15.10% and 20.65% relative to the conventional LSTM. Supplementary Table S1 additionally reports reductions of 34.88% and 27.08% relative to residual persistence.

With identical 145-variable historical inputs and two recurrent layers, the direct load-prediction LSTM achieved an RMSE of 132.102 kW, compared with 118.288 kW for the proposed model. This 10.46% reduction supports the combined contribution of residual-change learning and ETP-based load reconstruction under matched historical inputs.

Across five independent training runs, Physics-Enhanced RMSE and MAE were 117.271 ± 4.600 and 64.358 ± 2.800 kW (mean $\pm$ sample standard deviation), compared with 148.126 ± 9.687 and 86.794 ± 6.319 kW for the conventional LSTM. As shown in Fig. 6, mean errors decreased by 20.83% and 25.85%, with both metrics lower in every paired run. Across 9203 subsequent-month windows, Physics-Enhanced reduced the conventional LSTM's pooled mean RMSE and MAE by 41.35% and 46.48%. Residual persistence achieved lower pooled MAE, while Physics-Enhanced achieved slightly lower pooled RMSE. Supplementary Table S8 reports the monthly results.

For the original date partition, replaying the same capacity policy for all five Physics-Enhanced training runs, with the margin recalibrated from each run's validation errors at the fixed 0.93 quantile, gave public-opening coverage of $97.751 \pm 0.416\%$ and scheduled-operation coverage of $96.258 \pm 0.349\%$. Recommendation changes averaged 585.8 ± 7.3 over the 56 test days, with a range of 577–595. The following detailed comparisons use the reference run.

At the validation-selected quantile of 0.93, the margin was 130.972 kW. Relative to the 0.95 reference under the same post-closing rule, the public-opening excess-tier rate decreased from 22.00% to 17.46%, and mean excess capacity decreased from 543.10 to 509.95 kW, while coverage changed from 98.59% to 98.21%.

Across 7697 valid recommendations, capacity was lower than, equal to, or higher than the historical BMS capacity in 20.83%, 67.53%, and 11.64% of cases, respectively. There were 577 recommendation changes; load-coverage calculations used the 7691 records with valid measured targets.

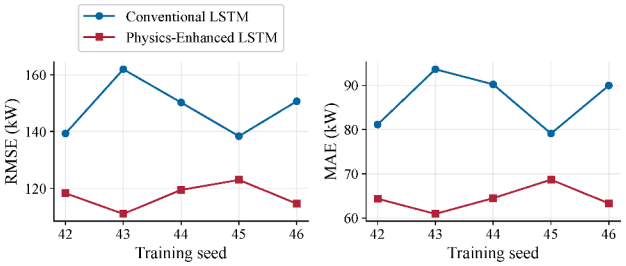


Fig. 6. 10-min-ahead forecast errors across five paired training runs, indexed by their random seeds. The data split and model configurations are fixed across runs.

Table 3
Capacity coverage and shortfall using the selected policy. Mean shortfall includes every valid row in the stated subset. Adjusted targets set the comparison load to zero at policy-defined OFF intervals; the other rows use measured loads.

Basis	Covered / total	Coverage (%)	Mean shortfall (kW)
Public opening	2638/2686	98.21	1.817
Scheduled operation	3995/4142	96.45	3.123
All measured targets	4193/7691	54.52	13.475
Adjusted targets	7601/7691	98.83	1.476

The continuous margin-adjusted forecast covered 91.38% of valid next-step loads. Discrete capacity selection adds headroom when the plant is recommended ON. Table 3 reports coverage after applying the full policy, separately for measured loads and policy-adjusted targets. Only 0.31% exceeded total installed capacity; all chillers were selected and the corresponding shortfall was recorded.

All 3425 OFF recommendations with valid measured targets occurred outside public opening hours. Historical logs recorded all chillers OFF at 3175 target timestamps and at least one chiller ON at 250. Mean,

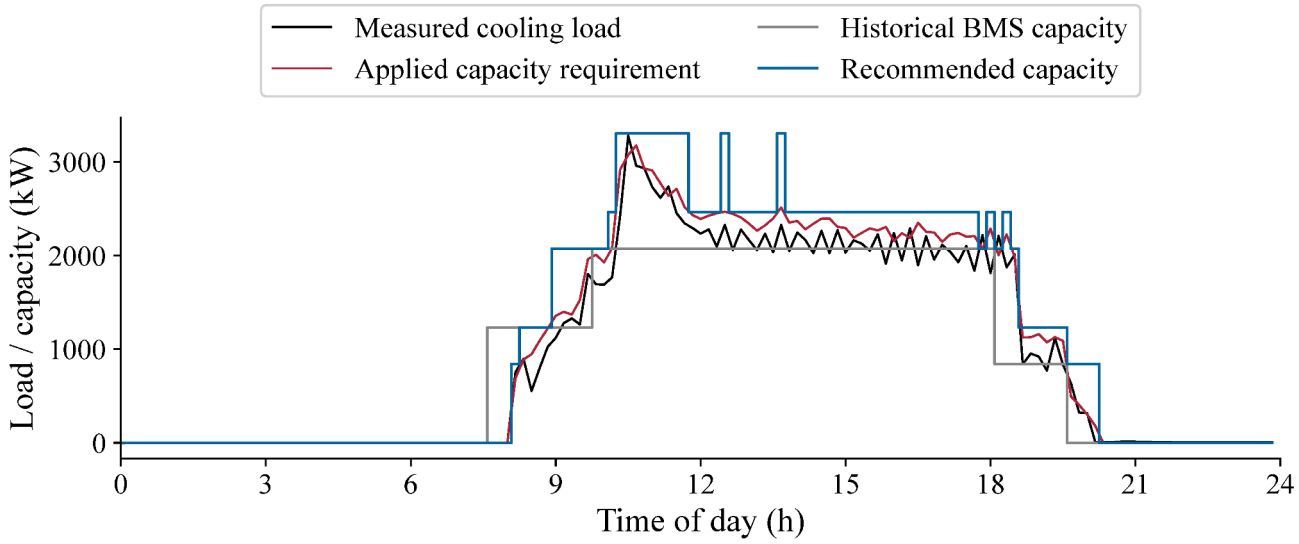


Fig. 7. 10-min-ahead capacity recommendations on 12 July 2024 using the selected margin and post-closing rule. The applied requirement includes the OFF gate.

Table 4
Prediction-to-decision comparison under the same policy. Coverage and load differences use public-opening records; changes use all valid recommendations.

Metric	Persistence	Physics-Enhanced
Coverage (%)	97.51	98.21
Mean excess capacity (kW)	549.91	509.95
Mean shortfall (kW)	6.810	1.817
Recommendation changes	780	577

P95, and maximum measured loads were 26.94, 57.37, and 490.74 kW. Seventeen loads exceeded the 74.141 kW threshold: 10 before public opening and 7 after closing. Thirteen of these timestamps had a recorded chiller ON. Table 3 distinguishes measured-load coverage from policy-adjusted diagnostics. During recorded all-chiller-OFF periods, nonzero water-side readings may reflect continued circulation, stored cooling, or measurement and timestamp effects. Mean return-air temperature rose by 0.375 °C over 60 min in 47 continuously recorded OFF events. Including subsequent restarts, the corresponding rises were 0.346 and 0.398 °C for 52 recommendation-linked origins and all 61 observed evening shutdowns. Supplementary Sections S4 and S8, Table S9, and Figs. S2–S3 detail event selection, restart frequencies, and sensor-level variation. These measurements characterize historical return-air responses; occupied-zone comfort under the recommended policy requires field evaluation.

Fig. 7 illustrates how the applied requirement is mapped to the available discrete capacity levels and compared with historical BMS capacity.

To evaluate the effect of learned residual changes on decisions, the latest-residual persistence forecast was replayed through the same recommendation policy. Each predictor used its own validation errors to calibrate a margin at the common quantile of 0.93; the persistence margin was 158.411 kW. The same 7697 decision origins were used. Recommended capacities differed at 706 origins. As shown in Table 4, residual-change learning improved public-opening coverage while reducing excess capacity, load shortfall, and recommendation changes.

4.3. Sensitivity to equivalent building parameters

Each of the 12 parameters in Table 1 was perturbed individually by ±20%, with floor count changed from five to four or six. The 24 perturbations were evaluated with fixed ETP coefficients and with coefficients recalibrated on the original training-period public-opening samples, giving 48 scenarios. Schedule, neural weights, and training scalers were fixed. Thermal states were recomputed chronologically before selecting test dates, with affected historical ETP features, residual anchors, and target baselines updated.

Table 5 reports errors on the common test windows. Floor-area perturbations produced the largest fixed-coefficient hybrid errors, 118.842–123.516 kW. Training-only recalibration narrowed hybrid RMSE to 118.275–118.309 kW, close to the 118.288 kW reference. Supplementary Fig. S1 illustrates this stability.

4.4. Evaluation with archived weather forecasts

Archived GFS temperature, humidity, and radiation forecasts from Open-Meteo's Previous Runs API were evaluated using its previous-day offset [24]. Under this convention, interpolated forecast values are available before the decision origin, although their valid timestamps may follow it. Network weights, ETP coefficients, and schedule were fixed; each margin was calibrated from validation errors at the 0.93 quantile.

Table 6 reports the original uncorrected comparisons. Replacing target-time weather increased reference-run RMSE from 118.288 to 176.644 kW; additionally using backward-aligned historical inputs gave 178.111 kW. Both cases achieved 95.41% scheduled-operation coverage.

An extended retrospective analysis used historical records to represent origin-time weather, assuming timely availability, and aligned archived forecast levels to these records while retaining forecast changes. Across five runs, mean RMSE and MAE were 117.065 and 64.398 kW, with both errors below the conventional LSTM in every paired run. Carrying the latest hourly weather forward gave similar accuracy. The conventional model excludes Open-Meteo inputs and is unchanged across protocols. The adjusted reference run achieved 96.38% scheduled-operation coverage and 509.26 kW mean public-opening excess capacity. Supplementary Section S6, Tables S5– S6, and Eq. (S5) provide the protocols, information timing, variability, and capacity results.

4.5. Daily capacity allocation

Both configurations met the policy-adjusted requirement in 6965 intervals, yielding downshift and upshift integrals of 138.920 and 28.609 MW·h of rated capacity and a net difference of 110.311 MW·h. The remaining 726 intervals comprised 636 with only historical capacity below the requirement, 26 with only recommended capacity below

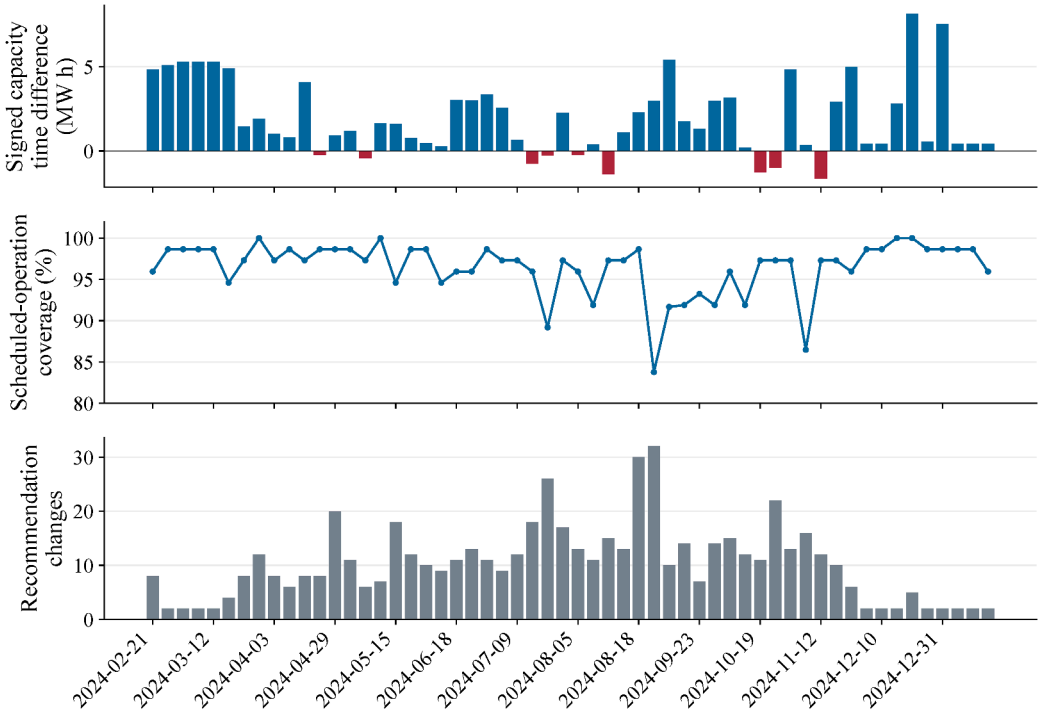


Fig. 8. Daily capacity-allocation results across all 56 test days: signed rated-capacity-time difference on comparable intervals, scheduled-operation capacity coverage, and recommendation changes. Positive differences indicate a larger downshift than upshift integral.

Table 5
Parameter sensitivity: RMSE (kW) for −20% / +20% perturbations with fixed coefficients, and ranges across the two perturbations after training-only recalibration. Neural weights remain fixed in both protocols.

Parameter	Fixed ETP	Fixed hybrid	Recalibrated ETP	Recalibrated hybrid
Floor area	356.347 / 385.977	118.842 / 123.516	313.670–313.680	118.284–118.293
Floor count	313.342 / 314.003	118.264 / 118.312	313.473–313.849	118.277–118.297
Floor height	313.104 / 314.393	118.231 / 118.352	313.458–313.919	118.275–118.304
Window-to-wall ratio	313.076 / 314.384	118.239 / 118.341	313.439–314.024	118.282–118.308
Wall U-value	313.686 / 313.673	118.282 / 118.294	313.611–313.746	118.286–118.291
Roof U-value	313.686 / 313.673	118.282 / 118.295	313.609–313.748	118.285–118.291
Window U-value	313.665 / 313.694	118.279 / 118.298	313.646–313.713	118.288–118.289
Solar heat-gain coefficient	313.095 / 314.367	118.245 / 118.336	313.435–314.022	118.282–118.309
Shading factor	313.095 / 314.367	118.245 / 118.336	313.435–314.022	118.282–118.309
Solar absorptance	313.665 / 313.696	118.286 / 118.291	313.629–313.729	118.287–118.290
Exterior convection	313.702 / 313.667	118.292 / 118.286	313.617–313.721	118.286–118.289
Effective heat capacity	313.727 / 313.632	118.288 / 118.289	313.638–313.722	118.288–118.289

Table 6
Historical-weather reference and uncorrected archived-weather comparisons with frozen reference-run weights.

Weather protocol	RMSE (kW)	MAE (kW)	Margin (kW)	Scheduled coverage (%)
Historical-weather reference	118.288	64.368	130.972	96.45
Archived day-ahead target weather	176.644	103.628	322.608	95.41
Archived target weather + backward-aligned history	178.111	104.630	323.558	95.41

it, and 64 with both below it. Across all 7691 valid intervals, the descriptive net capacity-time difference was 16.061 MW·h.

Considering only the comparable intervals defined in Eq. (23), the daily signed capacity-time difference was positive on 47 of the 56 test days and negative on the remaining nine days as reported in Fig. 8. The median daily difference was 1.386 MW·h, with values ranging from -1.660 to 8.139 MW·h. The median daily capacity coverage during scheduled operation was 97.30%, and the minimum was 83.78%. The number of changes in the recommended chiller combination ranged from 2 to 32 per day.

5. Conclusion and future work

This study combines a calibrated ETP baseline with LSTM predictions of residual changes to forecast cooling load and recommend discrete chiller capacities.

Under the fixed within-month partition and historical-weather reference, five independent training runs achieved a mean 10-min-ahead RMSE of 117.3 kW, 20.83% below the conventional LSTM, and mean public-opening capacity coverage of 97.8%. Validation-based margin selection reduced excess-capacity recommendations, while

residual-change learning reduced recommendation changes relative to persistence. Chronological validation across three subsequent test months retained the forecasting advantage over the conventional LSTM. Relative to residual persistence, Physics-Enhanced achieved slightly lower pooled RMSE, while persistence achieved lower MAE.

The analysis across all 56 test days reports daily differences between historical and recommended rated capacities, together with capacity coverage and recommendation changes. Forecast errors changed little under the tested parameter perturbations after the ETP coefficients were recalibrated using training data. The archived-weather tests assessed sensitivity to weather inputs, and the historical shutdown analysis measured changes in return-air temperature after shutdown. Future work should incorporate equipment efficiency and operational constraints into the recommendation method and evaluate thermal comfort using measurements from occupied zones.

CRediT authorship contribution statement

Zhenghao Gao: Writing – original draft; **Nan Wang:** Methodology; **Tong Wu:** Supervision; **Shang Yang:** Validation; **Jian Xiao:** Writing – review & editing; **Lunshu Chen:** Writing – review & editing; **Junyao Feng:** Writing – review & editing.

Declaration of competing interest

The author(s) declare(s) that the study does not involve humans nor animal subjects.

Acknowledgments

This paper is funded in part by the Guizhou Power Grid Co., Ltd. (GZKJXM20240010), in part by the Science and Technology Development Fund, Macao SAR (0089/2026/RIB1, 0050/2025/AIJ).

Data availability

The authors do not have permission to share data.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.enbuild.2026.118388.

References

[1] D. Ürge-Vorsatz, L.F. Cabeza, S. Serrano, C. Barreneche, K. Petrichenko, Heating and cooling energy trends and drivers in buildings, Renew. Sustain. Energy Rev. 41 (2015) 85–98.
[2] C. Fan, F. Xiao, Y. Zhao, A short-term building cooling load prediction method using deep learning algorithms, Appl. Energy 195 (2017) 222–233.
[3] D. Bai, S. Ma, L. Wu, K. Wang, Z. Zhou, Research on feature variable set optimization method for data-driven building cooling load prediction model, Buildings 15 (19) (2025) 3583.
[4] Z. Zhang, H. Hui, Y. Song, Microclimate-interactive energy management of building clusters: linking local meteorology effect with terminal operation efficiency, CSEE J. Power Energy Syst., Early Access (2026) 1–13, https://doi.org/10.17775/CSEEJPES.2025.04570.
[5] S.R. Thangavelu, A. Myat, A. Khambadkone, Energy optimization methodology of multi-chiller plant in commercial buildings, Energy 123 (2017) 64–76.
[6] Y. Xiang, L. Chen, Q. Wuyun, S. Yu, Y. Geng, H. Tang, et al., A comprehensive review of building-to-grid interaction, Energy Use 2 (1) (2026) 100043.
[7] Y. Liao, Y. Sun, G. Huang, Robustness analysis of chiller sequencing control, Energy Convers. Manage. 103 (2015) 180–190.
[8] Z. Zhang, H. Hui, Energy-saving potential assessment of building energy system considering climate impacts, in: 2023 IEEE Sustainable Power and Energy Conference (iSPEC), IEEE, 2023, pp. 1–5.
[9] M. Song, C. Gao, H. Yan, J. Yang, Thermal battery modeling of inverter air conditioning for demand response, IEEE Trans. Smart Grid 9 (6) (2018) 5522–5534.
[10] T. Qi, H. Hui, Y. Song, Exploring the feasible regulation region of building HVAC systems within hygro-thermal comfort, IEEE Trans. Smart Grid 16 (4) (2025) 2926–2939.
[11] Y. Yu, X. Si, C. Hu, J. Zhang, A review of recurrent neural networks: LSTM cells and network architectures, Neural Comput. 31 (7) (2019) 1235–1270.
[12] C. Zhou, Z. Fang, X. Xu, X. Zhang, Y. Ding, X. Jiang, Y. Ji, Using long short-term memory networks to predict energy consumption of air-conditioning systems, Sustain. Cities Soc. 55 (2020) 102000.
[13] S. Jiang, H. Hui, Energy consumption probabilistic prediction of HVAC systems in public buildings based on deep learning fusion model, in: 2024 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), 2024, pp. 71–76.
[14] Y. Wang, D. Gan, M. Sun, N. Zhang, Z. Lu, C. Kang, Probabilistic individual load forecasting using pinball loss guided LSTM, Appl. Energy 235 (2019) 10–20.
[15] C. Zhang, J. Li, Y. Zhao, T. Li, Q. Chen, X. Zhang, A hybrid deep learning-based method for short-term building energy load prediction combined with an interpretation process, Energy Build. 225 (2020) 110301.
[16] A. Li, F. Xiao, C. Zhang, C. Fan, Attention-based interpretable neural network for building cooling load prediction, Appl. Energy 299 (2021) 117238.
[17] Z. Wang, Y. Chen, Data-driven modeling of building thermal dynamics: methodology and state of the art, Energy Build. 203 (2019) 109405.
[18] ASHRAE, ANSI/ASHRAE Standard 30–2019: Method of Testing Liquid Chillers, ASHRAE, Atlanta, GA, USA, 2019.
[19] K. Hinkelman, S. Huang, J. Wang, J. Lian, W. Zuo, Enhancing the implementation of a first-order equivalent thermal parameter model to enable accurate and robust building thermal response prediction, in: Proceedings of Building Simulation 2019, 2019, pp. 1859–1865.
[20] Z. Ma, G. Jiang, J. Chen, Physics-informed ensemble learning with residual modeling for enhanced building energy prediction, Energy Build. 323 (2024) 114853.
[21] Z. Wang, T. Hong, M.A. Piette, Building thermal load prediction through shallow machine learning and deep learning, Appl. Energy 263 (2020) 114683.
[22] Z. Chen, J. Zhang, F. Xiao, K. Xu, Y. Chen, Development of a probabilistic cooling load prediction-based robust chiller sequencing strategy and its real-world implementation, Appl. Energy 382 (2025) 125213.
[23] C. Zhang, Y. Zhao, C. Fan, T. Li, X. Zhang, J. Li, A generic prediction interval estimation method for quantifying the uncertainties in ultra-short-term building cooling load prediction, Appl. Therm. Eng. 173 (2020) 115261.
[24] P. Zippenfenig, Open-Meteo.com Weather API, 2023, [Computer software], Zenodo, https://doi.org/10.5281/zenodo.7970649.